Chih-Kai Wang

B.S. Student | AI for Mathematics and Verifiable Reasoning

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EDUCATION

National Taipei University of Education | Expected 2028

B.S. student, Mathematics Education Division, Department of Mathematics and Information Education, National Taipei University of Education. Expected 2028.

RESEARCH INTERESTS

AI for Mathematics; automated theorem proving and autoformalization; verifier-guided research agents; evaluation and calibration; reproducible scientific computing; nonlinear differential equations and bifurcation theory.

RESEARCH DIRECTION

ClaimPromoteBench - Policy-Governed Evidence-State Transitions for Mathematical Research Agents | In development

- Designing a benchmark protocol for deciding whether evidence justifies a requested mathematical claim-state transition.
- Separating research, computation, formal-certificate, and independent-review states, with deterministic policy checks and counterfactual pairs.
- Current status: alpha research scaffold and protocol development; no model results or technical report have been published.

SELECTED OPEN-SOURCE WORK

ProofWeave Core v2 | Python, Lean 4

- Built model-independent infrastructure for parsing mathematical claims and proofs and checking constrained Lean 4 / Mathlib certificates.
- Keeps formal-certificate status, natural-language alignment, and claim lifecycle orthogonal; no model is called at runtime.
- Public identifiers: repository release v0.1.0; Core package 2.0.0.

RigorGraph | Python

- Built a local-first claim-evidence graph and deterministic audit workflow with provenance, dependency, snapshot, independence, and hash checks.
- Produces self-contained offline reports and preserves explicit truth boundaries. Public beta; package 1.0.1.

HonestCI | TypeScript

- Built a CLI and GitHub Action that checks fresh JUnit evidence and detects missing reports, zero-test runs, and suspicious test-count reductions.
- Publishes versioned evidence bundles without treating CI success as proof of program correctness. Stable version 1.0.4.

TECHNICAL SKILLS

Languages: Python, TypeScript | **Mathematics and research:** SymPy, LaTeX, reproducible computation

Formal methods: Lean 4 / Mathlib (developing) | **Engineering:** JSON Schema, GitHub Actions, cross-platform CI, deterministic validation

Last updated: August 2026